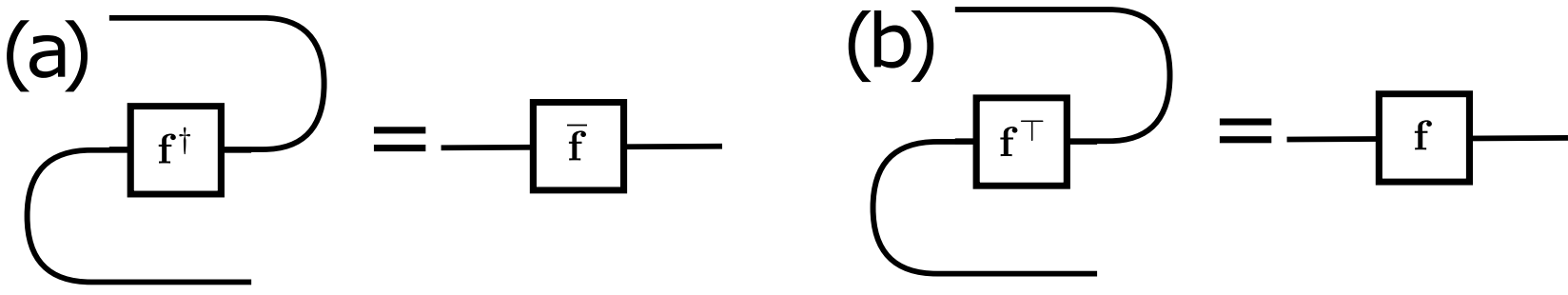
(a)
$\mathbf{f}^\dagger$
=
$\overline{\mathbf{f}}$
(b)
$\mathbf{f}^\top$
=
$\mathbf{f}$